Checkpoint 14: eigenfunction graph rigidity and slice-curvature compensation

1. Candidate-proof status

There is not yet a complete dimension-free proof, and no KLS conclusion is claimed at this checkpoint. The new work isolates one narrow second-order gate for a near-linear first eigenfunction, proves that gate on several genuinely dependent classes, and shows exactly why ordinary thin-shell information stops at a logarithm.

Let μ be isotropic and log-concave, and let

$$-L_\mu f = \lambda f, \qquad \mathbb{E}f = 0, \qquad \mathbb{E}f^2 = 1, \qquad \mathbb{E}|\nabla f|^2 = \lambda.$$

Write

$$a = \mathbb{E}[Xf(X)], \qquad \delta^2 = 1 - |a|^2, \qquad r = f - a \cdot x.$$

The tensor-extremizer route would make δ numerical and small if its convex-potential normal-cone term could be eliminated. The present checkpoint studies the independent question of whether such near-linearity already forces λ to be numerical.

2. Exact spectral and Bochner ledger

Isotropy and the weak eigenvalue equation give

$$\begin{aligned}
&\mathbb{E}r = 0, \qquad \mathbb{E}[Xr] = 0, \qquad \|r\|_2^2 = \delta^2, \\
&\mathbb{E}\nabla f = \lambda a, \qquad \mathbb{E}\nabla r = -(1-\lambda)a, \\
&\mathbb{E}|\nabla r|^2 = 1 - \lambda + (2\lambda - 1)\delta^2.
\end{aligned} \tag{2.1}$$

For smooth full-support measures, with the standard Reilly boundary term for convex supports, put

$$B_f = \mathbb{E}\|D^2 f\|_{\mathrm{HS}}^2, \qquad \mathcal{R}_V(f) \geq 0$$

for the sum of the potential-curvature and boundary-curvature terms. Bochner–Reilly and componentwise Poincaré yield the exact identities

$$B_f + \mathcal{R}_V(f) = \lambda^2, \tag{2.2}$$

$$\mathrm{Var}(\nabla f) = \lambda[1 - \lambda(1 - \delta^2)], \tag{2.3}$$

and the nonnegative defect decomposition

$$\boxed{[B_f - \lambda\,\mathrm{Var}(\nabla f)] + \mathcal{R}_V(f) = \lambda^3(1 - \delta^2).} \tag{2.4}$$

In particular $\|D^2 r\|_2 = \|D^2 f\|_2 \leq \lambda$. These formulas contain no unproved KLS input.

The tempting interpolation

$$\mathrm{Var}(\nabla f) \leq C\,\delta\,\|D^2 f\|_2$$

is false. The first degree-one Neumann mode on the isotropic Euclidean ball satisfies

$$\lambda = 1 - \frac{1}{n} + O(n^{-2}), \quad \delta = \frac{1}{n} + O(n^{-2}), \quad B_f = \frac{3}{n} + O(n^{-2}), \quad \mathrm{Var}(\nabla f) = \frac{1}{n} + O(n^{-2}),$$

so the ratio grows like $\sqrt{n/3}$.

3. The surviving mean-gradient gate

The narrower statement is

$$\boxed{|\mathbb{E}\nabla g| \le C(\|g\|_2 + \|D^2 g\|_2), \qquad \mathbb{E}g = 0, \quad \mathbb{E}[Xg] = 0.} \tag{MG}$$

Applied to r, it gives

$$(1-\lambda)\sqrt{1-\delta^2} \le C(\delta+\lambda), \tag{3.1}$$

which forces $\lambda \ge c(C) > 0$ once δ is a sufficiently small universal number.

There is a complete one-dimensional proof of the stronger Hessian-only form. If ν is centered, variance one, and log-concave, its one-dimensional Poincaré constant is at most 12. For $h \perp \{1, x\}$, setting $m = \int h'\,d\nu$ and applying Poincaré to h' and then to $h - mx$ gives

$$\left|\int h'\,d\nu\right| \le 12\left(\int (h'')^2 d\nu\right)^{1/2}. \tag{3.2}$$

For a product of such laws, the conditional projections $g_i(x_i) = \mathbb{E}[g \mid X_i = x_i]$ are orthogonal and

$$\begin{aligned} |\mathbb{E}\nabla g|^2 &\le 144 \sum_i \|g_i''\|_2^2 \\ &\le 144\,\mathbb{E}\|D^2 g\|_{\mathrm{HS}}^2. \end{aligned} \tag{3.3}$$

Thus there is no hidden dimension loss for products.

For a symmetric matrix B, let

$$q_B = X^T BX - \operatorname{tr} B, \qquad TB = \mathbb{E}[Xq_B], \qquad g_B = q_B - (TB)\cdot X.$$

If C is the covariance operator of centered quadratic forms and $\mathsf{S} = \mathsf{C} - T^*T$, then

$$\|g_B\|_2^2 = \langle B, \mathsf{S}B\rangle, \qquad \mathbb{E}\nabla g_B = -TB, \qquad D^2 g_B = 2B. \tag{3.4}$$

The Hessian-only quadratic gate is exactly

$$\sup_{|u|=1} \left\|\mathbb{E}[(u\cdot X)(XX^T - I)]\right\|_{\mathrm{HS}} \le C. \tag{3.5}$$

Projection thin shell gives, for every rank-k projection P,

$$|\operatorname{tr}(PM_u)| \le C\sqrt{k}.$$

Hence $|\lambda_j(M_u)| \le Cj^{-1/2}$ and only $\|M_u\|_{\mathrm{HS}} \le C\sqrt{\log(en)}$. The harmonic diagonal matrix shows that these rank-by-rank data alone cannot remove the logarithm.

Disintegrating along u also does not prove (MG). For $h(t) = \mathbb{E}[g \mid u\cdot X = t]$ and conditional score $s_t = V_t - \mathbb{E}_t V_t$,

$$\begin{aligned} h' &= \mathbb{E}_t g_t - \mathbb{E}_t(g s_t), \\ h'' &= \mathbb{E}_t g_{tt} - 2\mathbb{E}_t(g_t s_t) + \mathbb{E}_t[g(s_t^2 - \partial_t s_t)]. \end{aligned} \tag{3.6}$$

Prékopa controls only the scalar Fisher difference in (3.6); it does not control the signed score covariances. For hard supports these terms are moving-boundary fluxes.

4. Dependent classes where the gate closes

For the paired box wedge

$$K_a = \{(t,x,y) : |t| \le 1,\ |x_i| \le 1 + a_i t,\ |y_i| \le 1 - a_i t\},$$

put $S = \sum_i a_i^2$ and $v = \operatorname{Var} T$. The axial slice potential satisfies

$$W''(t) \ge 2S, \qquad \text{hence} \qquad vS \le \frac{1}{2}. \tag{4.1}$$

After exact covariance whitening, the only third coefficients are

$$\lambda_i = \frac{2a_i\sqrt{v}}{1 + a_i^2 v}, \qquad \sum_i \lambda_i^2 \le 2,$$

and therefore

$$\sup_{|u|=1} \|M_u\|_{\mathrm{HS}} \le 2. \tag{4.2}$$

The whitened law is the image of a product of isotropic one-dimensional log-concave laws under an explicit map F with $\|DF\|_{\mathrm{op}} \le 4$. Consequently

$$C_P(K_a^{\mathrm{iso}}) \le 192, \qquad |\mathbb{E}\nabla g| \le 192\|D^2 g\|_2 \quad (g \perp \mathrm{Aff}). \tag{4.3}$$

A genuinely one-sided wedge has

$$\|DF\|_{\mathrm{op}} \le \sqrt{28} + 3, \qquad C_P \le 444 + 72\sqrt{28} < 972, \tag{4.4}$$

and obeys the same type of Hessian-only estimate. Gaussian convolution preserves these dimension-free conclusions and produces smooth, full-support, genuinely dependent examples.

Several further exact classes close the quadratic gate:

- For every log-concave Dirichlet law with parameters $\alpha_i \ge 1$, $A = \sum_i \alpha_i$, put $q_i = \alpha_i / A$. A unit tangent direction is represented by coefficients a_i satisfying $\sum_i q_i a_i = 0$ and $\sum_i q_i a_i^2 = 1$, and

$$\|M_{u(a)}\|_{\mathrm{HS}}^2 = \frac{4(A+1)}{(A+2)^2}\left(\sum_i a_i^2 - 2\right) < 4. \tag{4.5}$$

  The constant 2 is asymptotically sharp.
- For cones over centrally symmetric bases,

$$\sup_{|u|=1} \|M_u\|_{\mathrm{HS}}^2 = \frac{4(n-1)(n+2)}{(n+3)^2} \longrightarrow 4. \tag{4.6}$$

- Affine box slices and affine positive-definite Gaussian covariance pencils have universal bounds. In both cases the second derivative of the axial log-volume charges precisely the squared Frobenius norm of the normalized covariance velocity.

These results rule out the most direct realization of the harmonic $j^{-1/2}$ spectrum. What remains uncontrolled is simultaneous non-affine shape change and noncommuting rotation of conditional covariances.

5. Function-specific localization and convex-set diagnostics

For a fixed normalized first eigenfunction, a hard localization driver with $C_t b_t = 0$, $b_t = \mathrm{Cov}_t(f, X)$, preserves its posterior mean pathwise. If

$$\mathcal{E}_t = \mathbb{E}_t |\nabla f|^2, \qquad R_T = \text{terminal survivor variance},$$

the rank-one defect theorem gives the exact ledger

$$1 \le \frac{192}{T}\lambda + 96\,\mathbb{E}[R_T \mathcal{E}_T]. \tag{5.1}$$

Under the energy-biased law $dQ = (\mathcal{E}_T/\lambda)dP$, a small gap therefore forces

$$\mathbb{E}_Q R_T \ge \frac{1}{96\lambda} - \frac{2}{T}. \tag{5.2}$$

The soft driver creates full curvature $\delta T I$ only at the complementary cost

$$1 - \mathbb{E} m_T^2 \le \frac{\lambda}{\delta T}. \tag{5.3}$$

Thus this scheme cannot preserve the signal and regularize its selected direction at a universal cost without a new adaptive-survivor theorem.

For a closed convex half-set A, $\mu(A) = 1/2$, put $p = \mu^+(A)$. Log-concavity of $F(t) = \mu(A + tB_2^n)$ gives the sharp scalar implication

$$\boxed{\mathbb{E}\, d(X, A) \ge \frac{2\log 2 - 1}{4p}.} \tag{5.4}$$

Conversely, the distance function shows $C_P(\mu) \ge (\mathbb{E} d(X, A))^2$. A universal upper bound in (5.4) is therefore itself KLS-strength. Thin shell handles centered balls and isotropy handles halfspaces, but neither controls arbitrary convex half-sets. Naive convexification of a general Cheeger witness fails even for the isotropic Laplace law.

6. Displacement audit and next round

For every balanced Brenier map, the midpoint Cayley potential q is globally 1-smooth and satisfies

$$\frac{1}{8}W_2^2 \le \mathbb{E}_\mu[\sigma q] \le \frac{3}{8}W_2^2, \qquad \int |\nabla q|^2 d\mu \le W_2^2. \tag{6.1}$$

A Poincaré bound restricted to all such generally nonconvex Cayley potentials is quantitatively equivalent to KLS. In the firm zero-strain subbranch q is convex. However, the outer-halves-to-middle-halves interval transport has zero entropy deficit and

$$\mathbb{E}_{\text{source}} r \ge \mathbb{E}_{\text{target}} r$$

for every convex r. Hence no prescribed-orientation signed convex certificate, even one chosen from the transport, can recover the positive quadratic displacement with an entropy-deficit error. The nonconvex Cayley potential is essential.

The next independent routes are:

1. exploit the special eigenfunction equation to prove (MG) without proving the full directional third-moment tensor bound;
2. extend slice-curvature compensation from commuting affine pencils to noncommuting conditional shape flows;
3. test whether the parallel coupling of log-affine tilts controls a low spectral mode rather than only the trace $\sum_i \|x_i\|_{H^{-1}}^2$;
4. seek a global laminar or incidence theorem for the cyclic zero-strain displacement branch.

Every item remains independent of the ordinary $\sqrt{\log n}$ covariance bootstrap. None is presently a completed KLS proof.